SARA BAKHTIARI

PhD Candidate in Mechanical Engineering | Computational Mechanics | Finite Element Modeling | Scientific Computing

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Eligible for U.S. internships through F-1 Curricular Practical Training (CPT), subject to university authorization.

PROFESSIONAL SUMMARY

PhD candidate in Mechanical Engineering at the University of Kentucky specializing in computational mechanics and biomechanics. Develops nonlinear finite-element models coupling passive constitutive behavior, active contraction, circulation, and remodeling. Experienced in Python, FEniCS/DOLFIN, MPI/PETSc, SLURM-based HPC, model validation, scientific post-processing, and machine learning for engineering data.

EDUCATION

PhD, Mechanical Engineering (Direct PhD) - University of Kentucky | Spring 2024-Present

Candidate since Apr 2026 | GPA 3.91/4.00 | Advisor: Jonathan F. Wenk, PhD

B.Sc., Mechanical Engineering - Iran University of Science and Technology (IUST) | 2016-2021

RESEARCH EXPERIENCE

Graduate Research Assistant - Computational Cardiac Mechanics | Spring 2024-Present

University of Kentucky - Lexington, KY

- Develop and extend MyoFE_H, a multiphysics finite-element framework coupling nonlinear left-ventricular mechanics, spatially distributed half-sarcomere kinetics, active contraction, and lumped-parameter circulation.
- Implement and evaluate Guccione- and Holzapfel-Ogden-based anisotropic constitutive laws to quantify effects on stress, pressure-volume behavior, deformation, and growth/remodeling predictions.
- Develop computational studies of myocardial fiber disarray and quantify effects on ventricular mechanics, mechanical efficiency, energetics, and baroreflex compensation.
- Develop an agent-based fibrosis model coupled with finite-element cardiac mechanics to study spatially evolving fibrosis and tissue-level mechanical adaptation.
- Run distributed simulations using Python, FEniCS/DOLFIN, MPI, PETSc, SLURM, and Linux HPC; develop validation/post-processing workflows for biaxial/shear tests, passive inflation, stress/strain fields, PV loops, energetics, and ParaView.

Machine Learning & Vision-Language Models for Electrospinning | 2026

- Apply pretrained and fine-tuned vision-language models to electrospinning videos to extract high-dimensional representations for fiber-diameter prediction.
- Evaluate embedding repeatability, learned bottleneck representations, feature variability, regression performance, and repeated cross-validation using Python/scikit-learn.

SELECTED RESEARCH PROJECTS

Passive Myocardial Constitutive Modeling: Compared Guccione and Holzapfel-Ogden formulations and quantified effects on ventricular stress, deformation, pressure-volume behavior, and growth/remodeling predictions.

Fiber Disarray, Efficiency & Baroreflex: Developed computational studies linking progressive fiber disorganization to ventricular mechanics, mechanical efficiency, ATP-related energetics, and baroreflex compensation.

Fibrosis & Remodeling: Develop an agent-based fibrosis model coupled with finite-element cardiac mechanics to study spatially evolving fibrosis and tissue-level mechanical adaptation.

PUBLICATIONS & RESEARCH OUTPUTS

Islam, R., Bakhtiari, S., Anugrah, G. G., Mehri, M., & Wenk, J. F. "Advancements in finite element modeling of myocardial infarction: a review." Biomechanics and Modeling in Mechanobiology, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0

Bakhtiari, S., Mehri, M., & Wenk, J. F. "Choice of Passive Myocardial Constitutive Law Drives Divergent Predictions of Left Ventricular Remodeling in Volume-Overload Conditions." Manuscript under review.

Bakhtiari, S., Mehri, M., & Wenk, J. F. "Effect of Myocardial Fiber Disarray on Left Ventricular Mechanical Efficiency and Baroreflex Compensation." Manuscript under review.

Selected presentation: "Choice of Passive Myocardial Constitutive Law Drives Differences in LV Mechanics and Growth Under Disease Conditions." Poster, BMES Annual Meeting, San Diego, CA, October 2025.

TEACHING EXPERIENCE

Teaching Assistant - Statics, University of Kentucky | Fall 2026

- Solved and explained undergraduate statics problems involving equilibrium equations, free-body diagrams, force/moment analysis, trusses, frames, and systematic engineering problem-solving methods.
- Supported classroom instruction, student questions, engagement, and class management; communicated technical concepts to students with different levels of understanding.
- Assisted with grading and examination supervision, evaluating engineering solution approaches and maintaining consistent exam procedures.

Teaching Assistant - Statics, Iran University of Science and Technology | 2021

- Supported undergraduate engineering-mechanics problem solving and helped students strengthen core statics concepts.

TECHNICAL SKILLS

Computational Mechanics: Nonlinear FEA, continuum mechanics, constitutive modeling, anisotropic soft-tissue mechanics, mixed incompressible formulations, Newton-Raphson methods

Scientific Computing: Python, MATLAB, NumPy, SciPy, Pandas, FEniCS/DOLFIN, UFL

HPC: MPI, mpi4py, PETSc, SLURM, Linux, distributed-memory computing, Singularity/container workflows

Modeling & Visualization: Cardiac mechanics, PV analysis, growth/remodeling, fibrosis, ParaView, Matplotlib, HDF5, XDMF